

Music Scope Pad

Video Selecting Interface by Natural Movement in VR Space

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Figure 1: Music Scope Pad

ABSTRACT

This paper describes a novel video selecting interface that enables us to select videos without having to click a mouse or touch a screen. Existing video players enable us to see and hear only one video at a time, and thus we have to play pieces individually to select the one we want to hear from numerous new videos such as music videos, which involves a large number of mouse and screen-touch operations. The main advantage of our video selecting interface is that it detects natural movements, such as head or hand movements when users are listening to sounds and they can focus on a particular sound source that they want to hear. By moving their head left or right, users can hear the source from a frontal position as the tablet detects changes in the direction they are facing. By putting their hand behind their ear, users can focus on a particular sound source.

CCS CONCEPTS

• Applied computing → Sound and music computing.

KEYWORDS

tablet application, video selection, music interface, head direction, spatial acoustics

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1 INTRODUCTION

Despite having access to a huge number of videos on the Internet, users can only watch and listen to a limited number due to the time it takes to evaluate many videos. Our goal was to construct an interface that would enable people to easily select videos such as musical videos that they had an affinity for from many unknown ones.

Our system, called Music Scope Pad, makes it possible to select a video source from the many available without the need for mouse clicks or other visual manipulations by detecting natural movements when users are listening to sounds and focusing on a particular sound source that they want to hear (Fig. 1).

2 RELATED WORK

Previous music retrieval methods that use queries such as similarity-based searching [12, 14, 16], text-based searching [13], or collaborative filtering-based searching [7, 15] are useful for narrowing the number of musical pieces, but after a list of rankings has been provided, users are required to listen to each song individually because no consideration has been given to finding songs one has an affinity for from the list.

Musicream [2] enables users to interact with many music collections by applying operations and providing functions for the order of play. Papipuun [6] and SmartMusicKIOSK [1] provides a music summary and enables quick listening in a manner similar to a stylus skipping on a scratched record. All these systems [1, 2, 6] enable users to save time by previewing songs from a list of rankings acquired from the results of music retrieval. However,

these systems also force users to listen to each song individually and involve many mouse operations.

Previously reported headphones with sensors to detect the direction users were facing or the location of the head could improve the sense of musical presence and create a realistic impression, but could not highlight parts according to their wishes [3, 8, 17]. It was difficult to clearly hear a particular musical source from many other sources with these headphones, including some that users may have preferred not to hear. There are music spatialization systems [10, 11] that allow users to control the localization of each part in real time through a graphical interface. However, it is difficult to control each musical source's location through this interface.

We previously proposed Music Scope Headphones that provided the aforementioned functions by using a digital compass that detects the orientation of the face and distance sensors that measure the distance between the hand and ear [4, 5]. As face-to-face exhibitions have become difficult due to social distancing in the wake of COVID-19, we investigated implementing similar functions in an app so that a wider audience can experience our demonstration by using a tablet.

3 MUSIC SCOPE PAD

The Music Scope Pad has two modes, video selection and video viewing. In the video selection mode, multiple videos are played back simultaneously in a virtual reality (VR) space (Fig. 1(a)). In the video viewing mode, one selected video is played (Fig. 1(b)). By installing the Music Scope Pad application on a tablet, users can experience the following.

Scoping function: Enable users to scope many video sources allocated in VR space by moving their heads left or right or by looking up or down. The function enables users to landscape videos and save time in previewing them.

Focusing function: Enables users to highlight a particular sound source they want to hear by placing their hand behind an ear. This function enables users to narrow the area in which sources are audible in VR space as if they were controlling the directivity of a microphone (Fig. 2(a)).

Switching function: Enables users to seamlessly change video sources in VR space through gestures such as holding their palm out. When the same video is highlighted for 3 seconds in the video selection mode, the system switches to the video viewing mode and the highlighted video is played. When the user holds his/her palm over the video viewing mode, the system switches to the video selection mode (Fig. 2(b)).

4 IMPLEMENTATION

We implemented the Music Scope Pad to enable to save time in previewing music video on the basis of the following two policies.

Reduced mouse and touch screen operations When selecting music with a computer or tablet, we generally have to play songs individually with many mouse or touch screen operations and these interruptions break the process of listening, similarly when a telephone is ringing. To solve this, we propose operations without mouse and touch screen operation that are achieved by detecting natural movements

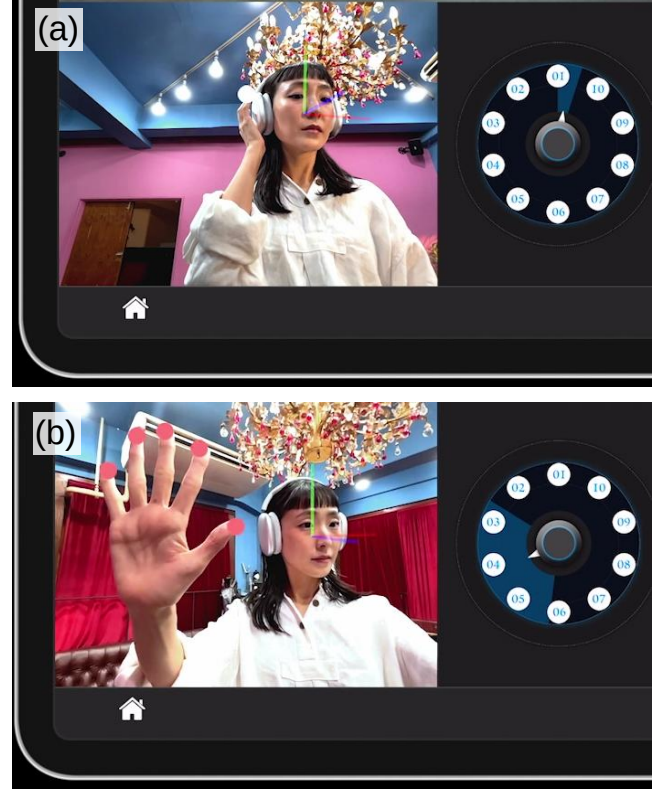


Figure 2: Lower Left Part of Screenshot

when listening to music and using these to control the system.

Easy preview of many songs We wanted to increase the number of opportunities for encountering unfamiliar musical videos in collections. However, the number of music videos that can be previewed is limited within a fixed amount of time. This is because the larger the number of videos to be previewed, the shorter the time to watch and listen to each video. To solve this, we propose a novel way of selecting music by playing many musical videos at the same time.

By using the Sound Scope Pad and arranging n video sounds in a VR space defined by the 3D sound library OpenAL [9], the sound of the videos arranged in the direction of the head can be heard in front. Here, let θ_n be the angle created from each of the n videos from the user's avatar. By changing the amplification rate h_n ($0 \leq h_n \leq 1$) in accordance with the orientation of the head, the sound of the video at an angle close to the front will be perceived as more emphasized. In the following, θ ($-\pi \leq \theta \leq \pi$) is the angle between the head direction and the position of the part, and δ ($0 \leq \delta \leq 1$) is the distance between the hand and ear (Fig. 3).

$$h_n^\theta = \begin{cases} 0 & h_n^\theta < 0 \\ h_n^\theta & 0 \leq h_n^\theta \end{cases}, \quad (1)$$

where

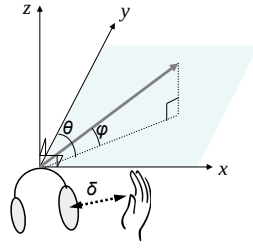


Figure 3: Direction θ , Angle of Elevation ϕ , and Distance δ

$$h_n^\theta = \begin{cases} 0 & \delta = 0 \\ 1 - (\alpha \cdot |\theta_n|) / (\pi \cdot \delta) & \delta > 0 \end{cases} \quad (2)$$

$\alpha (0 \leq \alpha \leq 1)$ is an adjustable parameter that sets the change in amplification rate when $\delta < 1$. When $\alpha = 0$, the amplification rate of each video does not change even if the hand is closer to the ear, but when $\alpha > 0$, the amplification rate decreases as the hand approaches the ear. At this time, because the amplification rate decreases more in the direction in which the user is not facing, the video sound in front can be heard relatively loudly.

5 CONCLUSION

We introduced the Music Scope Pad, an application that enables users to quickly select from many music videos their favorites. The Music Scope Pad application and an introductory video explaining how to use it can be downloaded at <https://gttm.jp/hamanaka/en/musicscopepad/> (Fig. 4).



Figure 4: Introduction Video of Music Scope Pad

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